Wk 5 Supervision Problems 1

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Permutations

1. Find the inverse of the following permutations and write your answer as a product of disjoint cycles.

   (i) (1 2 4)(3 5 7 6)

   (ii) (1 2)(2 4 3)(2 3 5)

2. Find all possible orders of elements in S_7. How many elements are there with order 2?

3. Let σ be a k-cycle. If m is a positive integer, prove that σ^m is a k-cycle if and only if k and m are coprime.

4. Prove that any subgroup of S_n either contains only even permutations, or exactly half the elements are even permutations.

Applying Lagrange's Theorem

1. Show that every group of order 55 contains an element of order 5 and an element of order 11.

2. Let G be a (possibly infinite) group with subgroups H and K such that K is also a subgroup of H. Suppose that $[G : H] = m$ and $[H : K] = n$. Show that $[G : K] = mn$.